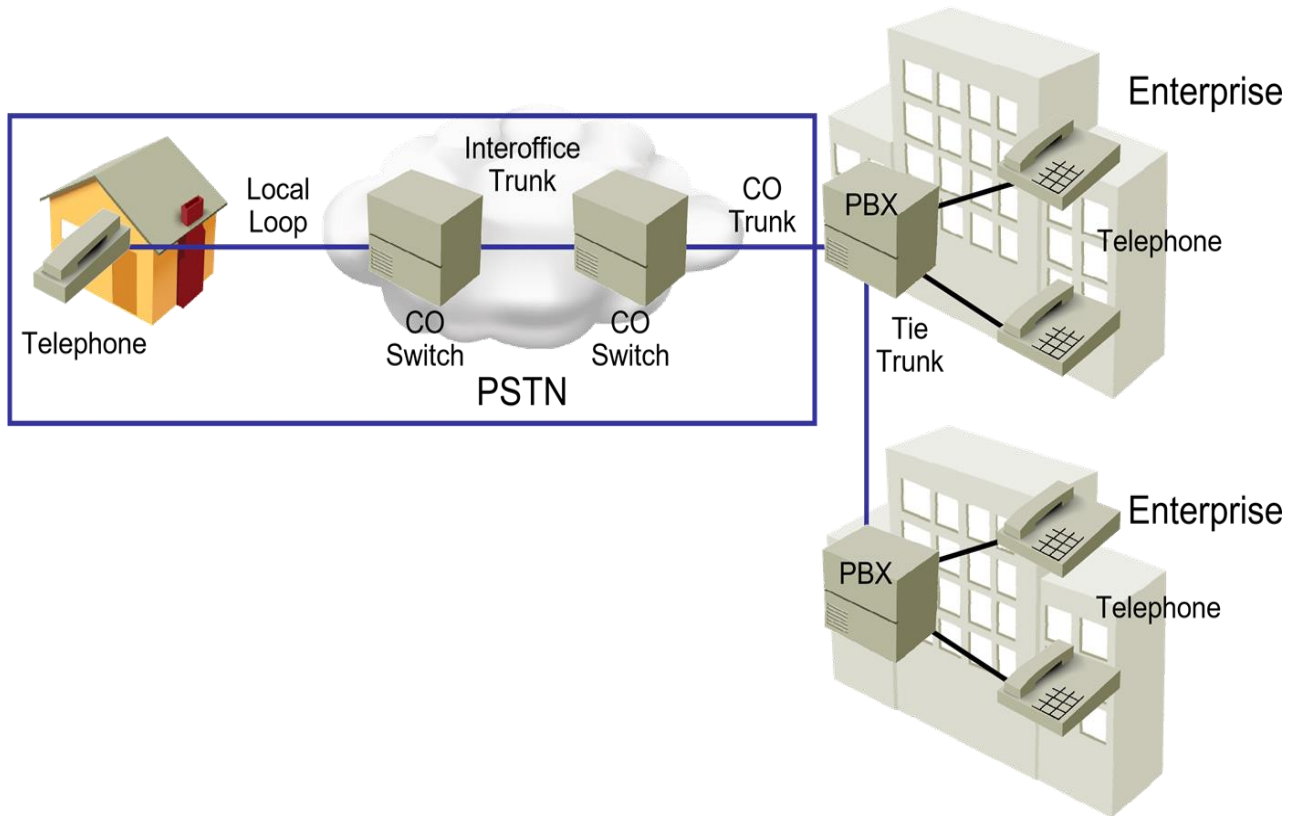
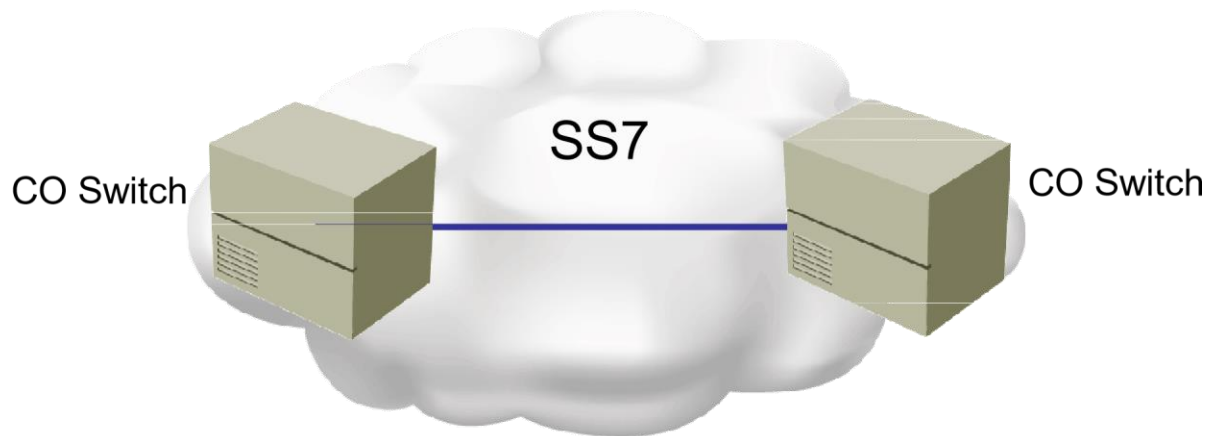


BASIC VoIP ARCHITECTURE

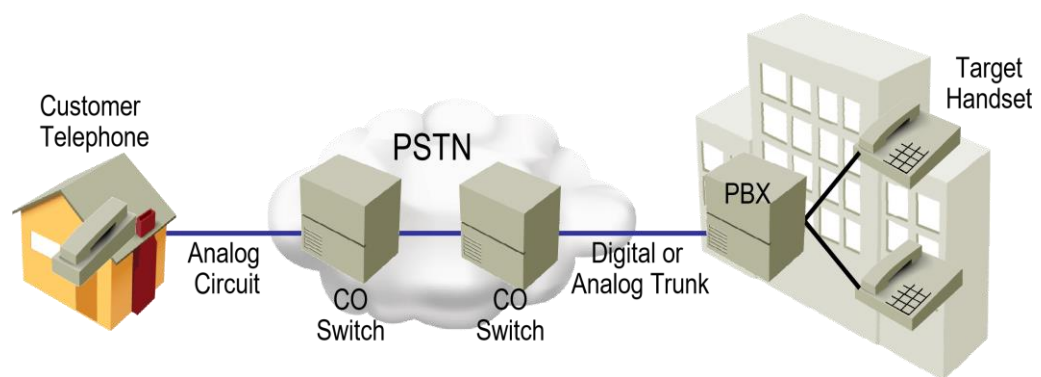
Public Switched Telephony Network



- There are three types of signaling used in a telephony network:
 - **Supervisory** signaling communicates the state of a telephony device.
 - **Address** signaling sends information about the digits dialed.
 - **Informational** signaling communicates the current state of the call.
- Signaling can be sent either **in-band** or **out-of-band**.
 - In-band signaling sends the signaling in the same communications channel as the voice.



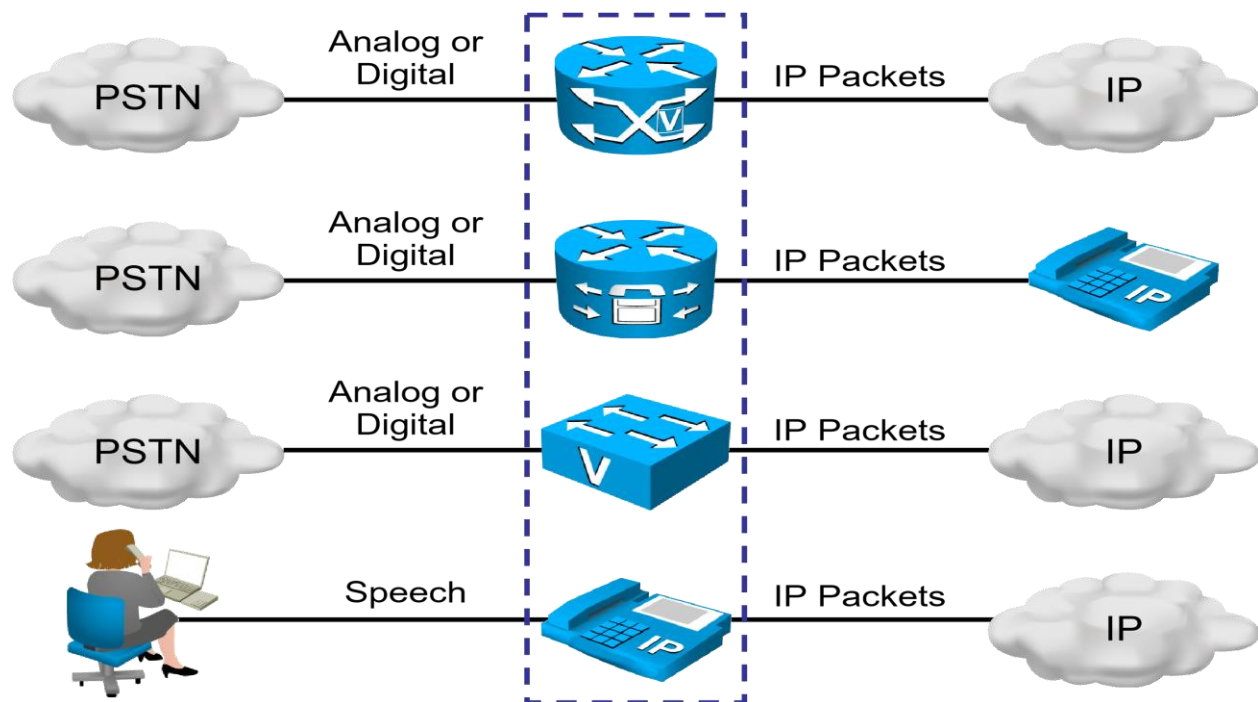
- SS7 functions:
 - Informational signaling
 - Call setup
 - Call routing
 - Call billing
 - Toll-free number resolution– Used out-of-band signaling²



1. Customer phone goes off hook creating a closed circuit.

2. The customer's CO switch detects that current is flowing and generates dial tone to the customer phone.
3. Either DTMF or pulse digits are dialed by the customer.
4. The CO switch collects the digits and performs an SS7 lookup to determine the destination CO switch.
5. Supervisory signaling indicates to the far-end analog or digital trunk that an inbound call has arrived.
6. The PBX determines which internal extension the call should go to and causes the target handset with that extension to ring.
7. Ringback is generated to the customer phone by their local CO switch.
8. The target handset goes off hook and a circuit is built end-to-end.

Voice Over Internet Protocol (VoIP)



Digital Signal Processor Chips

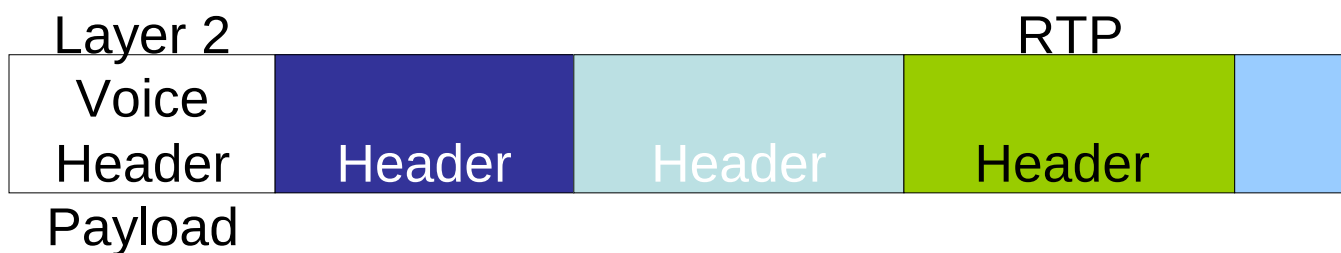


RTP Control Protocol

- Can be used to monitor the quality of the data distribution and provide control information
- Provides feedback on current network conditions
- Allows hosts that are involved in an RTP session to exchange information about monitoring and controlling the session:
 - Packet count

- Packet delay
 - Octet count
 - Packet loss
 - Jitter (variation in delay)
- Provides a separate flow from RTP for UDP transport use
 - Is paired with its RTP stream and uses the same port as the RTP stream plus 1 (odd-numbered port)

Packetization



- Packetization of voice is performed by DSP resources.
- The DSP packages voice samples or compressed voice into IP packets.
- The voice data is collected until the packet payload is full.

- The voice data is carried in the payload of RTP segments.
- RTP is encapsulated in a UDP segment, which is encapsulated in an IP packet.
- The IP packet is encapsulated into the Layer 2 format.

VoIP Signaling Protocols

- Signaling generates and monitors the call control information between two endpoints to:
 - Establish the connection
 - Monitor the connection
 - Release the connection
- The signaling protocol must pass **supervisory, informational, and address signaling.**
- Signaling protocols can be peer-to-peer or client/serverbased.
 - Peer-to-peer allows the endpoints to contain intelligence to place calls without assistance.

- Client/server puts the endpoint under the control of a centralized intelligence point.

The Media Gateway Control Protocol (MGCP) is a telecommunication protocol for signaling and call control in hybrid voice over IP (VoIP) and traditional telecommunication systems. It implements the media gateway control protocol architecture for controlling media gateways connected to the public switched telephone network (PSTN). The media gateways provide conversion of traditional electronic media to the Internet Protocol (IP) network. The protocol is a successor to the Simple Gateway Control Protocol (SGCP), which was developed by Bellcore and Cisco, and the Internet Protocol Device Control (IPDC).

The **Session Initiation Protocol** is a signaling protocol that enables the Voice Over Internet Protocol (VoIP) by defining the messages sent between endpoints and managing the actual elements of a call. SIP supports voice calls, video conferencing, instant messaging, and media distribution.

The **Signalling Connection Control Part (SCCP)** is a network layer protocol that provides extended routing, flow control, segmentation, connection-orientation, and error correction facilities in Signaling System 7 telecommunications networks.

VoIP Signaling Protocols Comparison

	Standards Body	Vendor Neutrality	Used on Gateways
H.323	ITU	Very Good	Yes
MGCP	IETF	Good	Yes
SIP	IETF	Basic	Yes
SCCP	None	Proprietary	Yes, limited

VoIP Disadvantages

- Requires stable internet connection.
- Jitter, latency, and dropped calls possible.
- No emergency services.